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The Dirichlet problem for Kolmogorov equations with measures as boundary conditions

Let a^{ij}, g^{ij}, b^i, h^i be Borel functions on $\mathbb{R}^d$, $A = (a^{ij}), G = (g^{ij})$, where the matrix A is positive definite. The elliptic operator

$$L_{A,b} f = \sum_{i,j} a^{ij} \partial_{x_i} \partial_{x_j} f + \sum_i b^i \partial_{x_i} f$$

$$= \operatorname{trace}(Af'') + \langle b, \nabla \rangle$$

generates the elliptic (stationary) Fokker-Planck-Kolmogorov equation

$$L^*_{A,b}\mu = 0,$$

with respect to a Borel probability measure μ on $\mathbb{R}^d$ is the identity

$$\int_{\mathbb{R}^d} L_{A,b} f(x)\, \mu(dx) = 0 \quad \forall\, f \in C_0^\infty(\mathbb{R}^d),$$

where it is assumed that a^{ij}, b^i are locally integrable with respect to μ.

Similarly this equation is defined on a domain $\Omega \subset \mathbb{R}^d$. How to introduce boundary conditions? There are papers studying boundary value problems with boundary conditions from $L^p(\partial\Omega)$:

P. Sjögren, On the adjoint of an elliptic linear differential operator and its potential theory, Ark. Mat. 11 (1973), 153–165.

P. Sjögren, Harmonic spaces associated with adjoints of linear elliptic operators, Ann. Inst. Fourier 25:3-4 (1975), 509–518.

L. Escauriaza, S. Montaner, Some remarks on the L^p regularity of second derivatives of solutions to non-divergence elliptic equations and the Dini condition, Rend. Lincei Mat. Appl. 28 (2017), 49–63.

Hongjie Dong, Seick Kim, On C^1, C^2, and weak type-$(1,1)$ estimates for linear elliptic operators, Comm. Partial Differ. Equ. 42:3 (2017), 417–435.
Hongjie Dong, L. Escauriaza, Seick Kim, On C^1, C^2, and weak type-$(1,1)$ estimates for linear elliptic operators: part II, Math. Ann. 370:1-2 (2018), 447–489.
Hongjie Dong, Dong-ha Kim, Seick Kim, Regular boundary points and the Dirichlet problem for elliptic equations in double divergence form, arXiv:2505.03137.
Seick Kim, Recent progress on second-order elliptic and parabolic equations in double divergence form. arXiv:2504.04892.

There are also papers on divergence form equations with measures as boundary conditions:
L. Verón, Elliptic equations involving measures. Stationary Partial Differential equations, pp. 593–712, V. 1, Handbook of Differential Equations, Elsevier, North Holland, Amsterdam, 2004.
M. Marcus, L. Véron, Nonlinear second order elliptic equations involving measures, de Gruyter, 2013.
But double divergence form equations with measures as boundary conditions have not been considered.
Set

$$\mathrm{div}^2 G = \sum_{i,j \leq d} \partial_{x_i} \partial_{x_j} g^{ij},$$

where the derivatives are understood in the sense of distributions.

Consider a more general double divergence form elliptic equation

$$\operatorname{div}^2(\varrho A) - \operatorname{div}(\varrho b) = \operatorname{div}^2 G - \operatorname{div} h, \qquad (1)$$

where a function $\varrho \in L^1_{loc}(\Omega)$ is called a solution if

$$a^{ij}\varrho,\ b^i\varrho,\ h^i, g^{ij} \in L^1_{loc}(\Omega)$$

and for all u one has

$$\int_\Omega L_{A,b}u(x)\varrho(x)\,dx = \int_\Omega L_{G,h}u(x)\,dx.$$

Our assumptions:

(H1) the functions a^{ij} and g^{ij} are continuous on Ω and there is a nondecreasing continuous function ω on $[0, +\infty)$ such that $\omega(0) = 0$ and
$\|A(x) - A(y)\| \leq \omega(|x - y|)$ for all $x, y \in \Omega$,

(H2) there is a number $\theta > 0$ such that
$\theta \mathrm{I} \leq A(x) \leq \theta^{-1}\mathrm{I}$ for all $x \in \Omega$,

(H3) for some $p > d$ we have

$$b^i \in L^p_{loc}(\Omega), \quad h^i \in L^1_{loc}(\Omega),$$

where $p' = p/(p-1)$.

A function $\varrho \in L^{p'}(D)$ is a solution to the Dirichlet problem

$$\begin{cases} \operatorname{div}^2(\varrho A) - \operatorname{div}(\varrho b) = \operatorname{div}^2 G - \operatorname{div} h, \quad x \in D, \\ \\ \varrho|_{\partial D} = \eta + \kappa\sigma_{d-1}, \end{cases}$$

where the function κ is defined by the formula

$$\kappa(x) = \frac{\langle G(x)\nu(x), \nu(x)\rangle}{\langle A(x)\nu(x), \nu(x)\rangle},$$

if for every $u \in W^{p,2}(D) \cap C^1(\overline{D})$ vanishing on ∂D

$$\int_D L_{A,b}u(x)\varrho(x)\,dx$$

$$= \int_D L_{G,h}u(x)dx + \int_{\partial D} \langle A(x)\nabla u(x), \nu(x)\rangle\,\eta(dx).$$

In case of smooth coefficients the second line holds as the equality for measures
$\varrho\,\sigma_{d-1} = (\varrho - \kappa)\,\sigma_{d-1} + \kappa\,\sigma_{d-1}$, i.e.,
$\eta = (\varrho - \kappa)\,\sigma_{d-1}$.
Let us see whence the boundary condition appears in the case of $A = \mathrm{I}$, $h = 0$, $G = 0$, and smooth b. Let ϱ be a solution to the equation $L^*_{\mathrm{I},b}\varrho = 0$ in Ω. Then $\varrho \in C^\infty(\Omega)$, hence for all $u \in C^2(\overline{D})$ vanishing on ∂D we have by the integration by parts formula

$$\int_D L_{\mathrm{I},b}u\varrho\,dx = \int_{\partial D}\langle\nabla u,\nu\rangle\varrho\,d\sigma_{d-1} - \int_D\langle\nabla u,\nabla\varrho\rangle\,dx$$
$$+\int_D\langle\nabla u,b\rangle\varrho\,dx = \int_{\partial D}\langle\nabla u,\nu\rangle\varrho\,d\sigma_{d-1},$$

since

$$\int_D \langle \nabla u, \nabla \varrho \rangle \, dx - \int_D \langle \nabla u, b \rangle \varrho \, dx$$

$$= - \int_D u[\Delta \varrho - \operatorname{div}(\varrho b)] \, dx = 0$$

due to the equation $\Delta \varrho - \operatorname{div}(\varrho b) = 0$. Thus, here the measure η is given by density ϱ with respect to the surface measure, i.e., is the restriction of the solution itself.

Theorem 1. Suppose that a function $\varrho \in L^{p'}_{loc}(\Omega)$ is a solution to equation (1) on Ω such that $\varrho(x) \geq 0$ for almost all $x \in \Omega$. Let D be a bounded domain with smooth boundary ∂D of class C^2 and $\overline{D} \subset \Omega$. Then there is a bounded Borel measure η on the surface ∂D such that the function ϱ is a solution to our Dirichlet problem on D with boundary condition $\eta + \kappa\sigma_{d-1}$ on ∂D.
If $\varrho \in L^{\infty}_{loc}(\Omega)$, then the assumption of nonnegativity of ϱ can be omitted, moreover, the corresponding measure η has a bounded density with respect to the surface measure σ_{d-1} on ∂D.

Theorem 2. For every finite Borel measure η on ∂D there exists a unique solution to our Dirichlet problem.